ZHIQI HUANG

FUKUOKA, JAPAN

EMAIL HOMEPAGE SCHOLAR GITHUB

RESEARCH PROFILE

- M.Phil. student working on geometry-aware generative modeling for interactive and embodied systems, using 3D geometry, PBR materials, and 3D Gaussian Splatting as structural intermediate representations for spatially consistent generation.
- Seeking RA and 2027 Ph.D. opportunities on 3D-aware video/world models, embodied simulation, controllable digital humans, and efficient multimodal 3D generation.

RESEARCH INTERESTS

- Geometry-aware generative modeling: geometry, depth, normal, material, and multi-view priors for 3D, video, and world models
- 3D-aware video/world models for embodied interaction, robot manipulation, AR/VR, and interactive simulation
- Simulation-ready assets and digital humans: relightable PBR materials, 3DGS avatars, controllable 3D objects, and consumer-GPU generation

EDUCATION

Waseda University — Apr. 2025–Mar. 2027 (expected)

M.Phil. in Information Architecture — Fukuoka, Japan

- English-taught program
- GPA: 3.8 / 4.0
- Research focus: geometry-aware generative modeling, 3D generation, and world models for embodied interaction

Sun Yat-sen University — 2017–2021

B.E. in Software Engineering — Guangzhou, China

- GPA: 3.7 / 4.0

PUBLICATIONS

SIE3D: Single-Image Expressive 3D Avatar Generation via Semantic Embedding and Perceptual Expression Loss — 2026

IEEE ICASSP 2026 (Accepted)

- First Author, Corresponding Author
- Authors: Zhiqi Huang, Dulongkai Cui, Jinglu Hu
- Introduced a multimodal conditioning scheme that fuses image identity embeddings with text semantic embeddings, plus a perceptual expression loss for expression-faithful 3DGS avatar generation.
- Targets controllable digital humans for games, VR, and interactive systems, while maintaining identity preservation and consumer-GPU practicality.
- Links: Project Page, Paper (arXiv:2509.24004), Code

MANUSCRIPTS UNDER REVIEW

First-author manuscript on geometry-aware PBR material generation from long text 2026

Currently under review

- First Author
- Uses long-text conditioning and geometry-gated attention with mesh-normal priors to ground material semantics on 3D surfaces.
- Generates relightable PBR maps for simulation-ready 3D assets and evaluates semantic alignment, reconstruction quality, and multi-view consistency.
- Designed around efficient, parameter-efficient adaptation and consumer-grade GPU inference.

INDUSTRY EXPERIENCE

4399 Games 2023–2024

Senior Graphics Engineer Guangzhou, China

- Led a 3–5 person rendering team and owned the rendering roadmap for *Era of Conquest* on mobile and PC.
- Drove graphics development for new projects including *Catch & Build: Land of Pals* across mobile, PC, and web.
- Built practical graphics workflows for shipped interactive games, shaping interest in controllable and efficient generative systems for games and simulation.

4399 Games 2021–2023

Graphics Engineer Guangzhou, China

- Built and optimized the real-time rendering pipeline for *Era of Conquest*, focusing on cross-platform shader optimization, PBR, and performance profiling.

TECHNICAL SKILLS

- Languages: C++, Python, C#, GLSL/HLSL
- Graphics & Engines: Vulkan, OpenGL, Unity, real-time rendering, physically based rendering (PBR)
- ML & 3D: PyTorch, 3D Gaussian Splatting, diffusion models, CLIP/LongCLIP, mesh processing, PBR material generation
- Research Practices: parameter-efficient fine-tuning, geometry-aware conditioning, multi-view rendering/evaluation, simulation-ready asset pipelines

LANGUAGES

- Chinese: Native (Mandarin and Cantonese)
- English: Professional working proficiency (TOEFL iBT: 90; English-taught master's program)

HONORS AND AWARDS

Outstanding Student Scholarship (Third Prize) 2018–2019

Sun Yat-sen University